

● RAG Learning

Required Practical Activities (PAGs)

OCR B GCSE Biology (J257) Twenty First Century Science — Practical Activity Groups

OCR B GCSE Biology requires students to complete at least eight practical activities covering eight Practical Activity Groups: PAG 1 Microscopy, PAG 2 Testing for biological molecules, PAG 3 Sampling techniques, PAG 4 Rates of enzyme-controlled reactions, PAG 5 Photosynthesis, PAG 6 Physiology, responses and respiration, PAG 7 Microbiological techniques and PAG 8 Transport in and out of cells. Schools can choose their own activity for each group, so the activities below are typical examples. Examiners test your understanding of these in the written papers. Each practical below shows the aim, key apparatus, method, expected results and how to handle the data and errors.

Exam Tip: When asked about a practical in the exam, always: (1) state the independent and dependent variables and at least 2 control variables; (2) describe how the dependent variable is measured; (3) explain how repeats and means improve precision; (4) suggest one source of error and how to minimise it.

PAG 1 — Microscopy: observing and drawing plant and animal cells

Aim: Use a light microscope to observe and draw plant and/or animal cells; calculate magnification.

Apparatus: Light microscope, slides, coverslips, onion epidermis (or cheek cells), iodine or methylene blue stain, mounted needle, ruler.

Method:

1. Peel a thin layer of onion epidermis (or scrape cells gently from inside the cheek).
2. Place flat on a slide with a drop of stain.
3. Lower a coverslip with a mounted needle to avoid air bubbles.

4. Place slide on the microscope stage; secure with clips.
5. Focus at the lowest power first; use coarse focus to bring stage close to lens, then move stage *away* while looking through eyepiece.
6. Switch to higher magnification; use only the fine focus.
7. Draw cells using sharp pencil; include a scale bar and total magnification.

Calculations: Total magnification = eyepiece × objective. Magnification = image size ÷ real size (use the same units throughout).

Hazards: sharp coverslip and slide edges; iodine stains skin and clothes — wear goggles and wash hands.

PAG 2 — Testing for biological molecules (food tests)

Aim: Identify the presence of carbohydrates, proteins and lipids in food samples.

Method (per food sample):

Molecule	Reagent	Procedure	Positive result
Reducing sugar (e.g. glucose)	Benedict's solution	Add sugar solution to Benedict's; heat in water bath 60–80 °C, 5 min	Brick-red precipitate
Starch	Iodine in potassium iodide	Add a few drops of iodine to the food	Blue-black colour
Protein	Biuret reagent	Add NaOH then dilute copper(II) sulfate (or Biuret)	Lilac/purple colour
Lipid	Ethanol + water (emulsion test)	Dissolve sample in ethanol; pour into water	White cloudy emulsion

Hazards: Benedict's and Biuret are irritants — wear goggles. Ethanol is highly flammable — keep away from naked flames. Use water baths not direct heat.

PAG 3 — Sampling techniques: quadrats and transects in the field

Aim: Investigate how the distribution of an organism (e.g. daisies) varies along an environmental gradient (e.g. light intensity from shaded to open ground).

Apparatus: Tape measure, quadrats (0.25 m²), light meter, soil moisture probe, identification key.

Method:

1. Lay a tape from shaded area into open ground.
2. At each marked interval (e.g. every 2 m) place the quadrat; record number of daisies and abiotic factors.
3. Repeat for several lines (transects) to improve reliability.

Calculations: Mean count per quadrat $\times$ area of habitat $\div$ area of quadrat = total estimated population. Plot abundance vs distance / abiotic factor.

Hazards: Slips/trips; sun exposure — wear suitable clothing.

PAG 4 — Rates of enzyme-controlled reactions: effect of pH on amylase

Aim: Investigate how pH affects the activity of amylase using starch and iodine.

Apparatus: Amylase solution, 1% starch solution, buffers pH 3–9, iodine solution in spotting tile, water bath at 30 °C, syringes, stopwatch.

Variables: IV — pH; DV — time for starch to be fully digested (no blue-black with iodine); CVs — temperature, enzyme concentration, starch concentration, volume.

Method:

1. Add 2 cm³ amylase + 1 cm³ buffer + 2 cm³ starch into a boiling tube; mix.
2. Place in 30 °C water bath; start stopwatch.
3. Every 30 s remove a drop and place on iodine in a spotting tile.
4. Stop the clock when iodine no longer turns blue-black.
5. Repeat for each pH; calculate mean.

Calculations: Rate = $1 \div \text{time (s}^{-1}\text{)}$. Plot rate vs pH; expect a peak at the optimum pH (≈ 7 for amylase).

Hazards: Iodine is irritant — wash off skin; buffers may be acidic/alkaline — handle with care.

PAG 5 — Photosynthesis: effect of light intensity on rate

Aim: Investigate the effect of light intensity on the rate of photosynthesis using pondweed (*Cabomba* or *Elodea*).

Apparatus: Pondweed, sodium hydrogen carbonate solution (CO_2 source), boiling tube, lamp, ruler, stopwatch.

Variables: IV — light intensity ($1/d^2$ where d = distance from lamp); DV — bubbles per minute (or volume of O_2 collected); CVs — temperature (use beaker of water as heat sink), CO_2 concentration, species/length of pondweed.

Method:

1. Cut a 5 cm piece of pondweed; place in tube with NaHCO_3 solution.
2. Set lamp at distance d ; allow 2 min to acclimatise.
3. Count bubbles released in 1 min; repeat 3 times and take mean.
4. Repeat at five different distances.

Calculations: Light intensity $\propto 1/d^2$ (inverse-square law). Plot mean bubbles min^{-1} vs $1/d^2$.

Hazards: hot lamp — do not touch; risk of electric shock if water spilled near plug.

PAG 5 — Respiration in yeast: effect of different substrates

Aim: Investigate how the respiratory substrate (e.g. glucose, fructose, sucrose, lactose, starch) affects the rate of respiration in yeast. OCR groups this with PAG 5 in the specification.

Apparatus: Yeast suspension, solutions of different sugars at the same concentration, methylene blue indicator (or a gas syringe / upturned measuring cylinder to collect CO₂), water bath, stopwatch.

Variables: IV — type of sugar; DV — time for methylene blue to decolourise (it is reduced as it accepts hydrogen from respiring yeast) OR volume of CO₂ produced per minute; CVs — temperature, yeast concentration, sugar concentration, volumes.

Method:

1. Warm the yeast suspension and each sugar solution separately in a water bath (e.g. 35 °C) for 5 minutes.
2. Mix yeast + one sugar solution in a tube; add a few drops of methylene blue.
3. Keep the tube in the water bath; do not shake (shaking re-oxidises the dye).
4. Time how long the indicator takes to lose its blue colour.
5. Repeat for each sugar, and include a control with water instead of sugar; take means of 3 repeats.

Calculations: Rate = 1/time (or cm³ CO₂ per minute). Plot a bar chart of mean rate for each sugar, since the IV is categoric. Yeast respire glucose fastest; sucrose is usually used quickly too, while sugars the yeast has no enzyme for (e.g. lactose) or cannot break down quickly (starch) give little or no respiration.

PAG 6 — Physiology and responses: investigating reaction times

Aim: Measure reaction time using the dropped-ruler test and investigate the effect of caffeine (or a distraction).

Apparatus: 30 cm ruler, partner, calculator, optional caffeine drink (with informed consent).

Method:

1. Subject sits with hand at the table edge, thumb and index finger 5 cm apart.
2. Tester holds the ruler vertically with 0 cm at the bottom of subject's fingers.
3. Tester drops the ruler without warning; subject catches it as quickly as possible.

4. Record the distance fallen (cm) above the thumb.
5. Repeat 5 times; calculate mean.
6. Use a conversion table or $t = \sqrt{(2d/g)}$ to convert distance into time.

Calculations: Reaction time $t = \sqrt{(2d/g)}$ where $g = 9.81 \text{ m s}^{-2}$. Convert d to metres.

Hazards: Caffeine — ensure informed consent and no relevant medical issues.

PAG 6 — Physiology and responses: effect of exercise on pulse rate and recovery

Aim: Investigate how exercise affects pulse rate (and breathing rate), and how long each takes to return to its resting value.

Apparatus: Stopwatch, step or bench (or a fixed space for jogging on the spot), metronome, optional heart-rate monitor.

Variables: IV — exercise (or its intensity/duration); DV — pulse rate (beats per minute) and recovery time; CVs — type and length of exercise, same person or people of similar age and fitness, same body position when measuring.

Method:

1. Sit quietly for 5 minutes, then count the pulse at the wrist (radial artery) for 30 s and double it to get beats per minute. Repeat to get a steady resting value.
2. Do a set exercise, e.g. step up and down a bench in time with a metronome for 3 minutes.
3. Measure the pulse immediately after stopping, then every minute until it is back to the resting value.
4. Record recovery time = time from stopping until the resting rate is reached.
5. Repeat with several volunteers (or at different exercise intensities) and compare means.

Calculations: Plot pulse rate against time; the rise shows the heart delivering more oxygen and glucose to the muscles for respiration. A fitter person usually has a lower resting rate and a shorter recovery time.

Hazards: Only volunteers who are fit to exercise should take part (check for asthma or heart conditions); stop if dizzy or unwell.

PAG 7 — Microbiological techniques: effect of antiseptics or antibiotics on bacterial growth

Aim: Investigate the effect of an antiseptic or antibiotic on bacterial growth using a clear zone of inhibition.

Apparatus: Sterile agar plates, broth culture of safe bacterium (e.g. *Bacillus subtilis* or *Micrococcus luteus*), sterile cotton swabs, paper discs, antiseptic/antibiotic solutions, control disc soaked in distilled water, forceps, Bunsen burner, ruler.

Method:

1. Use aseptic technique throughout: pass forceps and inoculating loop through a Bunsen flame; lift agar lid only briefly.
2. Spread bacterial culture evenly over the agar with a sterile swab.
3. Place paper discs (different antiseptics + control) onto the agar with sterile forceps.
4. Tape the lid (do not seal completely — risk of anaerobic growth).
5. Incubate at 25 °C for 48 hours (UK schools should not exceed 25 °C to avoid growing pathogens).
6. Measure the diameter of the clear "zone of inhibition" around each disc with a ruler.

Calculations: Zone area = $\pi \times (\text{radius})^2$; larger area = more effective inhibition.

Hazards: microbiological hazard — work aseptically; do not open plates after incubation; the teacher sterilises used plates in an autoclave (121 °C) before disposal.

PAG 8 — Transport in and out of cells: osmosis in potato tissue

Aim: Investigate the effect of sucrose concentration on the mass of potato tissue (osmosis).

Apparatus: Cork borer, potato, scalpel, balance, sucrose solutions $0.0\text{--}1.0\text{ mol dm}^{-3}$, boiling tubes, paper towels.

Method:

1. Cut potato cylinders of equal length; blot dry; record initial mass.
2. Place one cylinder in each sucrose concentration.
3. Leave for 30 min, then remove, blot, and record final mass.
4. Calculate % **change in mass** for each concentration.

Calculations: % change = (final – initial) / initial $\times 100$. Plot % change vs concentration; the line crosses 0 % at the concentration equal to the cell sap (isotonic point).

Hazards: sharp scalpel; clean up spillages.

Supporting practical (not a required PAG)

Rate of decay using milk and lipase (not an OCR PAG activity)

This practical is an AQA required practical, not one of OCR's eight PAGs. It is kept here because J257 separate-science content asks you to explain how temperature affects the rate of decomposition and to calculate rates of decay.

Aim: Investigate factors affecting the decay of milk or food using the pH fall caused by lipase breaking down milk fat as a model for decay.

Apparatus: Milk, lipase solution, sodium carbonate (alkali), phenolphthalein indicator, water baths.

Variables: IV — temperature; DV — time taken for indicator to lose its pink colour (acid produced as fats are broken down to fatty acids); CVs — milk volume, lipase concentration, indicator volume.

Method:

1. Mix milk + sodium carbonate + phenolphthalein in a tube — solution turns pink (alkali).

2. Add lipase; start stopwatch.
3. Stop the clock when pink colour disappears (pH falls below 8.3).
4. Repeat at 10 °C, 20 °C, 30 °C, 40 °C, 50 °C.

Calculations: Rate = 1 / time. Plot rate vs temperature; expect a peak at the lipase's optimum (≈ 35 °C) and denaturation above.

Practical skills you must master

Skill	Tip
Variables	Always state IV, DV and at least 2 control variables.
Repeats and means	Repeats expose anomalies; means smooth random error.
Reliability vs accuracy	Reliable = same result if repeated; accurate = close to true value.
Anomalies	Identify, ignore in mean calculation, suggest cause.
Errors	Random (variation in counts) vs systematic (mis-zeroed balance) — give an example.
Graphs	IV on x-axis, DV on y-axis. Use sensible scales; line/curve of best fit; label axes with units.
Conclusion	State trend, support with figures from data, refer back to the hypothesis.

Exam Tip: When asked "how would you improve this experiment?" — give a SPECIFIC suggestion (e.g. "use a colorimeter instead of judging colour by eye to remove subjective error"), not generic ones like "use better equipment".